AI reinvention made real

Accenture Innovation Challenge 2026

Instructions

Referenceslide–Remove before submission

1. Use the given template for your idea submission.

1. Follow file naming format: Team name_Idea Name.pptx

3. Ensure the spell check is done before submitting

4. Use standard Arial font, alignment and relevant images (as required)

Team details

Adithya Vijaya Raghavan

College: Indian Institute of Technology Madras
Stream: Naval Architecture and Ocean Engineering (B.Tech)
Year of graduation: 2027

Ugine Mercy

College: Indian Institute of Technology Madras
Stream: Metallurgical and Materials Engineering (B.Tech + M.Tech)
Year of graduation: 2028

Describe the problem statement (200 words)

DigitalTwin.ai

Problem Statement

Automobile assembly lines are highly intricate and are vulnerable to malfunctions. These disruptions create unwanted bottlenecks and early defects, leading to a waste of expensive resources and lower productivity. By implementing continuous monitoring and early problem resolution, plants can significantly increase their efficiency and overall yield.

Issue #1

What are things we need to model?

- Cycle times
- Transfer Times
- Buffer Capacities
- Kinematics of Robotic Arms
- Failure Probabilities
- Integration of PLC, sensor data
- Worker Fatigue

Issue #2

How to handle where there are no sensors:

- **Algorithmic Imputation:** Deduce the happenings in one station by looking at the previous and next stations.
- **Computer Vision:** Instead of sensors, we could mount inexpensive cameras and train them to identify defects and log times.

Our Goal

To develop a digital twin prototype of a vehicle assembly line which would :

- **Detect bottlenecks** and **optimise**.
- **Identify** early **defects** and **notify**.
- **Maximise** our **output**.

DigitalTwin

A live, data-driven virtual model capable of replicating the entire assembly line. It **continuously updates** itself in real-time based on live conditions, **accurately predicting** exactly how the physical line will behave.

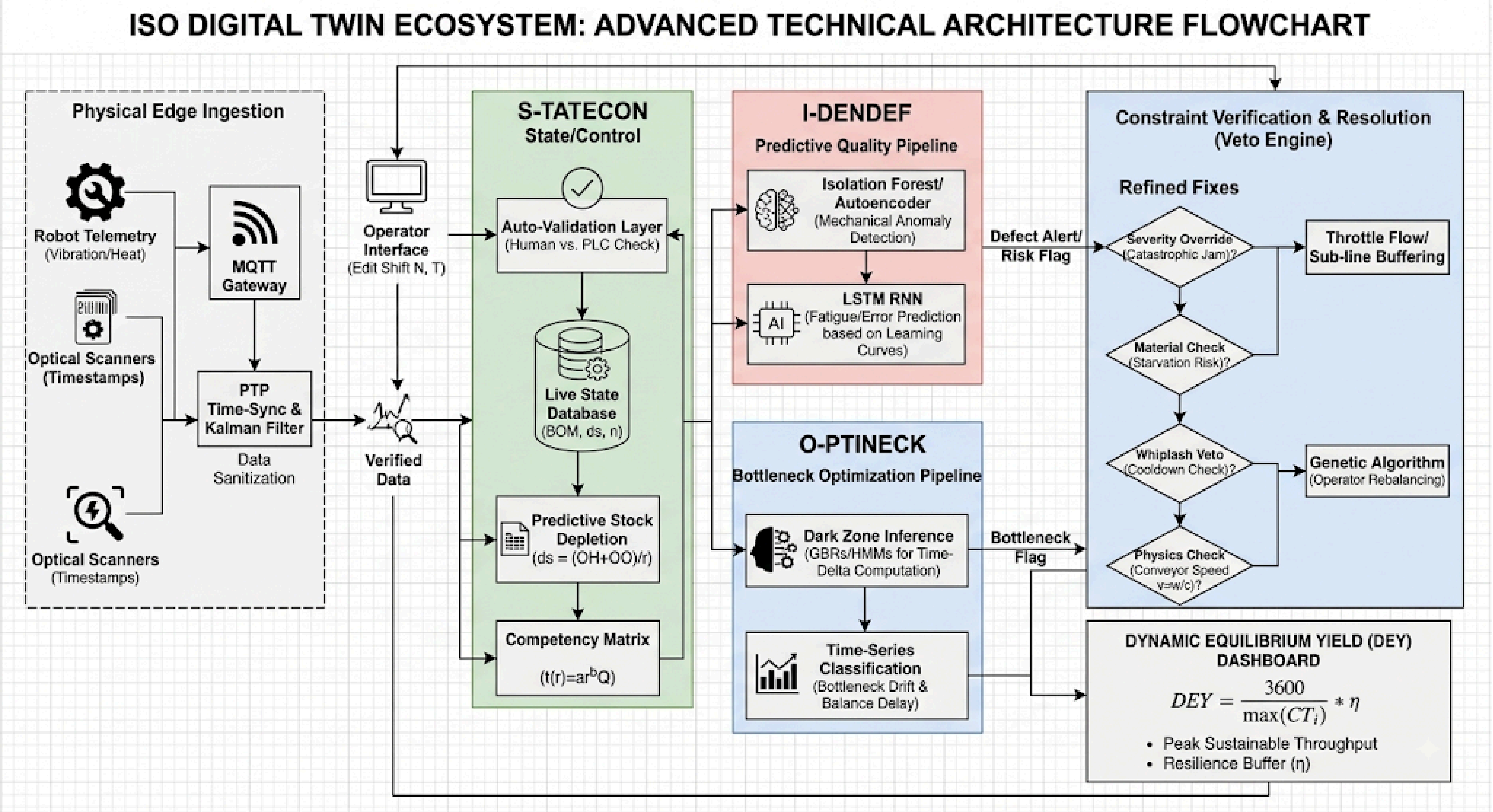
Could it be modeled?

Absolutely.

Our Solution:

ISO

Proposed solution (200 words)



Proposed solution (200 words)

ISO

Our DigitalTwin product which models,analyse and predict the physical assembly line with ease. It has three main architectures inbuilt namely, I-DENDEF, S-TATECON, O-PTINECK. All three tackle assist and tackle the problems together.

I-DENDEF

I-DENDEF is a predictive quality pipeline that intercepts defects at critical "Prime Elements" using edge-deployed AI on tool telemetry.

- **Mechanism:** Autoencoders flag mechanical vibration anomalies, while LSTMs estimate operator fatigue via Wright's learning curve.
- **Critical Loopholes:** Unsupervised AI risks "Concept Drift" from degrading sensors, requiring static physics anchors. Furthermore, predicting dynamic physiological exhaustion using a static learning curve is mathematically flawed and risks hallucinating defect alerts.

S-TATECON

S-TATECON (Synchronized Tracking of Assets, Telemetry, and Employees) is the central control engine and single source of truth for the ISO digital twin.

- **What it does**: It manages the live factory state (inventory, workers, and bill of materials) and allows managers to run instant "what-if" simulations.
- **How it does it**: It calculates theoretical cycle time and days' supply of stock, using an auto-validation layer to cross-reference human inputs against physical machine data.

O-PTINECK

O-PTINECK is a proactive flow optimizer designed to predict and prevent assembly line jams.

- **What it does**: It manages macro-flow by detecting micro-delays before they escalate into physical bottlenecks.
- **How it does it:** It tracks balance delay and utilises a "Dark Zone" inference engine to deduce task times at uninstrumented stations using upstream and downstream checkpoint time deltas.

Video

Thank you